Yifei Dong

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Education

2025, summer **Duke University**, *Visiting Scholar*, Robot Dexterity Lab, with Prof. Xianyi Cheng

2023 - 2026 **KTH Royal Institute of Technology**, *PhD Candidate*, Robotics, Computer Science

2019 - 2022 **ETH Zurich**, *M.Sc.*, Mechanical Engineering

2015 - 2019 **Shanghai Jiao Tong University**, *B.Sc.*, Mechanical Engineering
Tsien Hsue-shen Program & Zhiyuan Honors Program of Engineering

Employment

2023 - 2026 **Robotics, Perception and Learning, KTH**, *Stockholm*
Doctoral Student, with Prof. Florian T. Pokorny. PhD defense date: Dec. 1, 2026

2022 **Idiap, EPFL**, *Martigny*, *Research Assistant*, with Dr. Sylvain Calinon
Code contribution: Robotics Code From Scratch

2022 **F&P Robotics**, *Zürich*, *Software Engineer Intern*
Motion planning of door opening and human-robot interaction for a mobile service robot.

2018 **NIO Inc.**, *Shanghai*, *Software Engineer Intern*
LiDAR simulation of an autonomous vehicle in various traffic scenes.

Publications

Peer-reviewed Journals and Conference Proceedings (* indicates equal contribution):

[J1] Yifei Dong, Xianyi Cheng, Florian T. Pokorny, *Characterizing Manipulation Robustness through Energy Margin and Caging Analysis*, IEEE Robotics and Automation Letters (**RA-L**), 2024, *Paper*

[J2] Haofei Lu, Yifei Dong, Zehang Weng, Florian T. Pokorny, Jens Lundell, Danica Kragic, *Grasping a Handful: Sequential Multi-Object Dexterous Grasp Generation*, IEEE Robotics and Automation Letters (**RA-L**), 2025, *Paper* | *Website*

[C1] Kei Ikemura*, Yifei Dong*, Florian T. Pokorny, *Latent Diffeomorphic Co-Design of End-Effectors for Deformable and Fragile Object Manipulation*, Proceedings of Robotics: Science and Systems (**RSS**), 2026, *Paper*

[C2] Yifei Dong*, Yan Zhang*, Sylvain Calinon, Florian T. Pokorny, *Robustness-Aware Tool Selection and Manipulation Planning with Learned Energy-Informed Guidance*, IEEE International Conference on Robotics and Automation (**ICRA**), 2026, *Paper*

[C3] Yifei Dong*, Shaohang Han*, Werner Friedl, Rafael I. Cabral Muchacho, Máximo A. Roa, Jana Tumova, and Florian T. Pokorny, *CageCoOpt: Enhancing Manipulation Robustness through Caging-Guided Morphology and Policy Co-Optimization*, IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025, *Paper*

[C4] Yifei Dong, Florian T. Pokorny, *Quasi-static Soft Fixture Analysis of Rigid and Deformable Objects*, IEEE International Conference on Robotics and Automation (**ICRA**), 2024, *Paper*

[C5] David Cáceres Domínguez*, ..., Yifei Dong* et al. *The First WARA Robotics Mobile Manipulation Challenge–Lessons Learned*, European Conference on Mobile Robots, 2025, *Paper* | *Video*

[C6] Yue Chen, Yifei Sun, Lu Wang, Fangkai Yang, Pu Zhao, Minjie Hong, Yifei Dong, Minghua He, Nan Hu, Jianjin Zhang, Zhiwei Dai, Yuefeng Zhan, Weihao Han, Hao Sun, Qingwei Lin, Weiwei Deng, Feng Sun, Qi Zhang, Saravan Rajmohan, Dongmei Zhang, *DUET: Joint Exploration of User Item Profiles in Recommendation System*, Annual Meeting of the Association for Computational Linguistics (**ACL**), findings, 2026, *Paper*

Under Review and Tech Report:

[1] Yifei Dong, Zhanyi Sun, Lujie Yang, Manuel Baum, Shuran Song, Florian T. Pokorny, Xianyi Cheng, *Robustness of Robotic Manipulation: Foundations and Frontiers*, under review (IJRR), 2026, *Preprint*

[2] Kei Ikemura*, Yifei Dong*, David Blanco-Mulero, Alberta Longhini, Li Chen, Florian T. Pokorny, *Sim-to-Real Gentle Manipulation of Deformable and Fragile Objects with Stress-Guided Reinforcement Learning*, under review (ICRA), 2026, *Preprint*

[3] Yan Zhang, Yiming Li, Yifei Dong, Florian T. Pokorny, Sylvain Calinon, *Physics-Informed Eikonal Caging for Whole-Arm Manipulation Planning*, tech report, 2026, *Preprint*

[4] David Blanco-Mulero, Yifei Dong, Julia Borras, Florian T. Pokorny, Carme Torras, *T-DOM: A Taxonomy for Robotic Manipulation of Deformable Objects*, under review (RAS), 2025, *Preprint* | *Website*

[5] Christoffer K. Øhrstrøm, Rafael I. C. Muchacho, Yifei Dong, Filippos Moumtzidellis, Ronja Güldenring, Florian T. Pokorny, and Lazaros Nalpantidis, *Parabolic Position Encoding: Vision-Centric, Principled, Extrapolatable, General*, under review (NeurIPS), 2026, *Preprint*

[6] Haofei Lu, Hongjia Liu, Yifei Dong, Florian T. Pokorny, Jens Lundell, Danica Kragic, *MoDex: A Diffusion Policy for Sequential Multi-Object Dexterous Grasping*, under review (CoRL), 2026

Academic Activities

Invited Talk:

2026, July — Invited talk at Prof. Xiaolong Wang's group, University of California San Diego

2025, Feb. — Guest lecture at Duke University's graduate course, Robotic Manipulation, 2025 Spring

2024, Dec. — Invited talk at Prof. Joel Burdick's group, California Institute of Technology

Paper Review:

2026 — Associate Editor, IEEE International Conference on Robotics and Automation (ICRA)

2026 — Reviewer for Transaction on Robotics (T-RO) – 1 paper reviewed

2026 — Reviewer for The International Journal of Robotics Research (IJRR) – 1 paper reviewed

2024 - now — Reviewer for IEEE Robotics and Automation Letters (RA-L) – 9 papers reviewed

2026 — Reviewer for Transactions on Machine Learning Research (TMLR) – 1 paper reviewed

2026 — Reviewer for Conference on Robot Learning (CoRL) – 1 paper reviewed

2024 - now — Reviewer for IEEE International Conference on Robotics and Automation (ICRA) – 4 papers reviewed

2024 - now — Reviewer for IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) – 3 papers reviewed

2024 — Reviewer for International Symposium on Robotics Research (ISRR) – 1 paper reviewed

Supervision and Teaching:

2024 - 2026 — Supervised master/undergraduate students:

Kei Ikemura (next: PhD, KTH), Hongjia Liu (next: PhD, UCSD), Zhenyuan Liang (next: Research engineer, Stealth, USA), Li Chen, Yuxiao Zhu, Gawtam C. Ramesh

2024, 2025 — Teaching assistant in Introduction to Robotics, DD2410, KTH

Academic Service:

2026 — Main organizer of ICRA 2026 Workshop on manipulation robustness, Vienna | *Website*

References

Prof. Florian T. Pokorny, Professor, KTH Royal Institute of Technology, Sweden. Email: fpokorny@kth.se

Prof. Xianyi Cheng, Assistant Professor, Duke University, USA. Email: xianyi.cheng@duke.edu

Prof. Danica Kragic, Professor, KTH Royal Institute of Technology, Sweden. Email: dani@kth.se

Dr. Sylvain Calinon, Senior Research Scientist, Idiap Research Institute; Lecturer, EPFL, Switzerland. Email: sylvain.calinon@epfl.ch

Experiences

2024 **Mobile Manipulation Challenge, ABB**, *Västerås, Sweden*
Team member of KTH: Custom gripper co-design for multi-task manipulation | Slides

2021 **Robotic Systems Lab, ETH**, *Zürich*, *Graduate Researcher*, with Prof. Marco Hutter
Master thesis: Mobile door state estimation and parameter identification | Thesis
Semester thesis: Contact-implicit MPC for mobile manipulation | Video | Thesis

2020 **Computer Vision and Geometry Lab & Microsoft**, *Zürich*
Course project: Object mesh reconstruction using RGB-D cameras. | Thesis | Git | Video

2019 **SAIC Motor**, *Shanghai*
Bachelor thesis: Strategy optimization of autonomous emergency braking

2018 **Institute of Design and Control Engineering, SJTU**, *Shanghai*
3-DOF plum processing system for surface-curving and core-deprivation. | Video

2017 **Institute of Intelligent Vehicle, SJTU**, *Shanghai*
Structural design and locomotion formulation of a snake robot. | Video

Awards & Grants

2024	Travel Grant from the Karl Engvers Foundation
2023	European Commission Project *SoftEnable* Grant (Funding of Doctoral Program)
2021, 2022	ETH Scholarship for International Students
2019	Excellence Award, BAIC Automobile Invitational Tournament
2018	Tan Kah Kee Invention Award
2016, 2018	General Motors Scholarship
2017	Huawei Scholarship
2016, 2017	Zhiyuan College Honors Scholarship

Skills

Robot	Franka Panda, Allegro hand, Ufactory Xarm 7, ANYmal legged robot, etc.
Programming	Python, C++, MATLAB, C, C#
Tools	ROS/ROS2, Genesis, Mujoco, Pybullet, Isaac Gym, Git, Docker, Blender, PyTorch, Codex, Claude, Simulink, Unity3D, CAD (ANSYS, SolidWorks, CATIA, NX Unigraphics)
Language	English (TOEFL 109/120), Chinese (native), German (Goethe B1)